with intercellular spaces. The pith extends between the vascular bundles. are called primary pith rays or primary medullary rays. Function of the pith is **storage of food.**

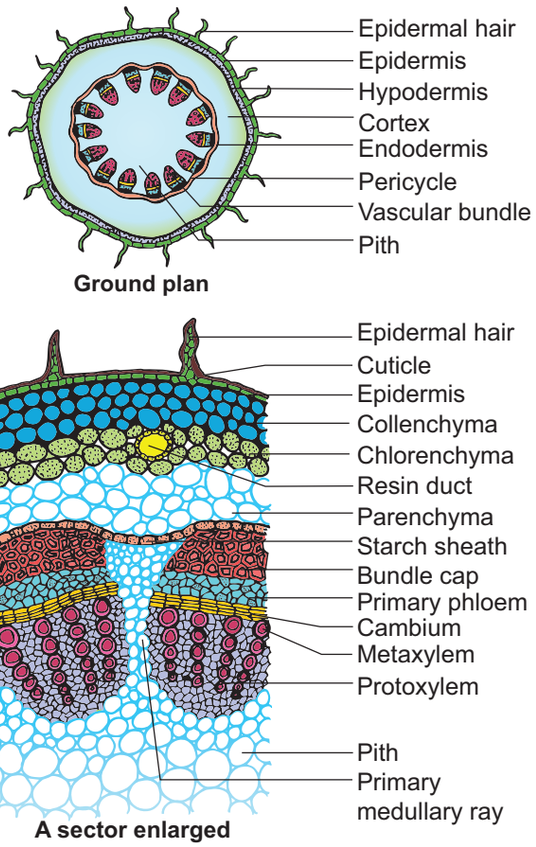


Figure 9.18: T. S of Dicot Stem (Sunflower stem)

Primary Structure of Monocot Stem-Maize Stem

Epidermis

It is the outermost layer of the stem. It is made up of single layer of tightly packed parenchymatous cells. Their outer walls are covered with thick cuticle. The continuity of this layer may be broken here and there by the presence of a few stomata. There are no epidermal outgrowths.

Hypodermis

A few layer of sclerenchymatous cells lying below the epidermis constitute the hypodermis. This layer gives mechanical strength to the plant. It is interrupted here and there by chlorenchyma cells.

Ground Tissue

There is no distinction into cortex, endodermis, pericycle and pith. The entire mass of parenchyma cells lying inner to the hypodermis forms the ground tissue.

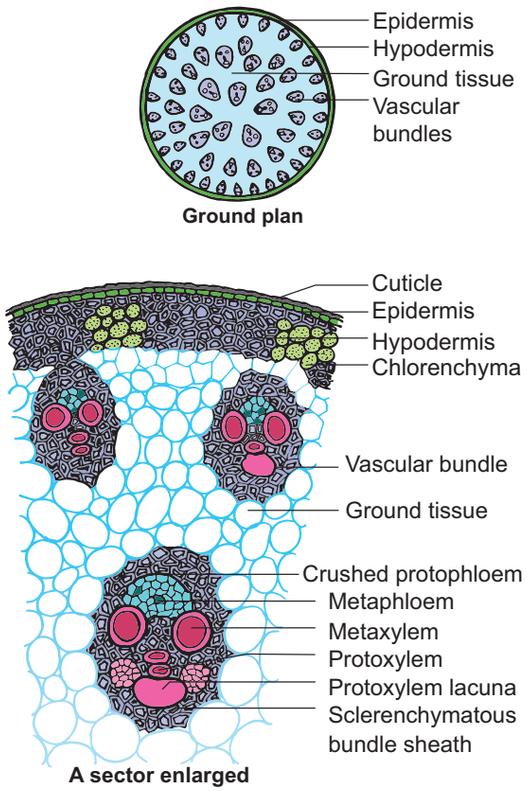


Figure 9.19: T.S. of Monocot stem (Maize stem)

The cell wall is made up of **cellulose**. The cells contain reserve food material like **starch**. The cells of the ground tissue next to the hypodermis are smaller in size, polygonal in shape and compactly arranged.

Towards the centre, the cells are loosely arranged, rounded in shape and bigger in size. The vascular bundles lie embedded in this tissue. The ground tissue stores food and performs gaseous exchange.

Vascular Bundles

Vascular bundles are **scattered (atactostele)** in the parenchymatous ground tissue. Each vascular bundle is surrounded by a sheath of sclerenchymatous fibres called **bundle sheath**. The vascular bundles are **conjoint, collateral, endarch** and **closed**.Vascular

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